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॥वसुधैव कुटुम्बकम्॥

Department of Computer Science and Engineering
DevOps Lab - L3

Lab Assignment No: 15	
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PRN	23070122170
Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Basic Infrastructure as Code (IaC) with Terraform

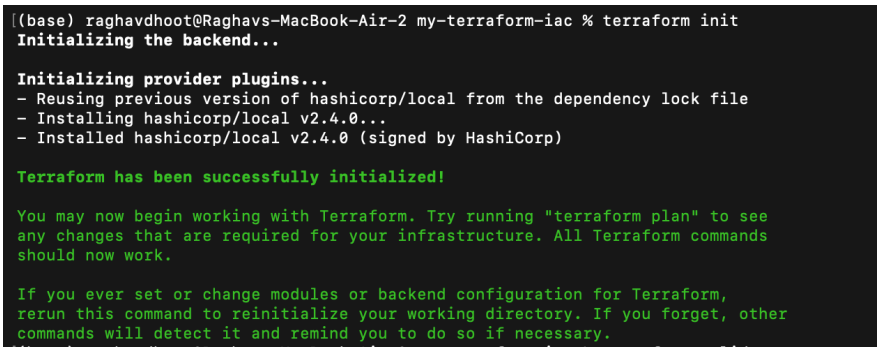
Implementation & Theory	`(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % terraform --version`
	`Terraform v1.16.5`
	`on darwin_arm64`
	
	Terraform installed — Installed via Homebrew (hashicorp/tap/terraform); confirmed with terraform --version.
	
	Project directory created — Set up Assignment - 15/my-terraform-iac as the working directory for this exercise.

```
terraform {
  required_providers {
    local = {
      source  = "hashicorp/local"
      version = "2.4.0"
    }
  }
}

provider "local" {
}

resource "local_file" "my_example_file" {
  filename = "example.txt"
  content  = "Hello, Terraform IaC!"
}
```

main.tf written — Defined a Terraform configuration using the hashicorp/local provider (pinned to version 2.4.0) and a local_file resource that creates example.txt with the content "Hello, Terraform IaC!"



terraform init — Downloaded and installed the local provider, initializing the working directory. Confirmed with "Terraform has been successfully initialized!"

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-terraform-iac % terraform validate
Success! The configuration is valid.

(base) raghavdhoot@Raghavs-MacBook-Air-2 my-terraform-iac % terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # local_file.my_example_file will be created
  + resource "local_file" "my_example_file" {
      + content              = "This content has been updated by Terraform!"
      + content_base64sha256 = (known after apply)
      + content_base64sha512 = (known after apply)
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "example.txt"
      + id                   = (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.
```

terraform plan — Generated an execution plan showing local_file.my_example_file would be created — Plan: 1 to add, 0 to change, 0 to destroy.

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-terraform-iac % terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # local_file.my_example_file will be created
  + resource "local_file" "my_example_file" {
      + content              = "Hello, Terraform IaC!"
      + content_base64sha256 = (known after apply)
      + content_base64sha512 = (known after apply)
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "example.txt"
      + id                   = (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

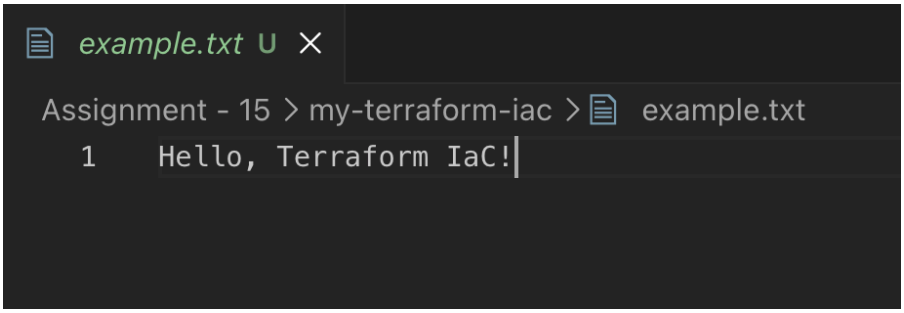
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.my_example_file: Creating...
local_file.my_example_file: Creation complete after 0s [id=67c458d620445706ef3f2d7483fd5f652f7ba55b]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

terraform apply — Applied the plan (confirmed with yes); Terraform created example.txt on disk. Verified with cat example.txt showing the expected content.




```
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-terraform-iac % terraform validate
terraform plan
Success! The configuration is valid.

local_file.my_example_file: Refreshing state... [id=67c458d620445706ef3f2d7483fd5f652f7ba55b]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.my_example_file must be replaced
-/+ resource "local_file" "my_example_file" {
      ~ content              = "Hello, Terraform IaC!" -> "This content has been updated by Terraform!" # forces replacement
      ~ content_base64sha256 = "NLXRlAVHreTKPhSWCELrt43/Dr6ErYjuUte0WBmBJUM=" -> (known after apply)
      ~ content_base64sha512 = "/HzlQIYMWvySOb0sTvaKq3qaHiLhLzVQVFSeSUbBSho4juvj56bTwBDeLcpof9zTTZfMKzGmN+Dem+D2npJtTQ==" -> (known after app
      ~ content_md5          = "ea9b237e8d0755ddbf752dfd68fa922f" -> (known after apply)
      ~ content_sha1         = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
      ~ content_sha256       = "34b5d1940547ade4ca3e14960842ebb78dff0ebe84ad88ee52d7b4c019bc2543" -> (known after apply)
      ~ content_sha512       = "fc7ce540860c32fc9239bd2c4ef68aab7a9a1e22e12f355054549e494c14a1a388eebe3e7a6d3c010de2dca687fdcd34d97cc2b31a63
      ~ id                   = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
        # (3 unchanged attributes hidden)
    }

Plan: 1 to add, 0 to change, 1 to destroy.
```

```
(base) raghavdhoot@Raghavs-MacBook-Air-2 my-terraform-iac % terraform apply
local_file.my_example_file: Refreshing state... [id=67c458d620445706ef3f2d7483fd5f652f7ba55b]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.my_example_file must be replaced
-/+ resource "local_file" "my_example_file" {
      ~ content              = "Hello, Terraform IaC!" -> "This content has been updated by Terraform!" # forces replacement
      ~ content_base64sha256 = "NLXRlAVHreTKPhSWCELrt43/Dr6ErYjuUte0wBm3JUM=" -> (known after apply)
      ~ content_base64sha512 = "/HzlQIYMWvySOb8sTvaKq3qaHiLhLzVQVFSaSUb8Shq4juvj56bTWBDeLcpof9zTTZfMKzGmN+Dem+D2npJtTQ==" -> (known after apply)
      ~ content_md5          = "ea9b237e8d0755ddbf792dfd68fe922f" -> (known after apply)
      ~ content_sha1         = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
      ~ content_sha256       = "34b6d1940547ade4ca3e14960842ebb78dff0ebe84adb8ee52d7b4c019bc2543" -> (known after apply)
      ~ content_sha512       = "fc7ce54086e8c32fc9239bd2c4ef68aab7a9e1e22e12f3558545494e4946c14a1a388eebe3e7a6d3c010de2dca687fdcd34d97cc2b31a63
      ~ id                   = "67c458d620445706ef3f2d7483fd5f652f7ba55b" -> (known after apply)
        # (3 unchanged attributes hidden)
    }

Plan: 1 to add, 0 to change, 1 to destroy.

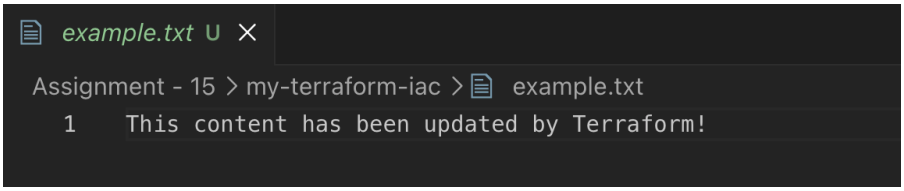
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

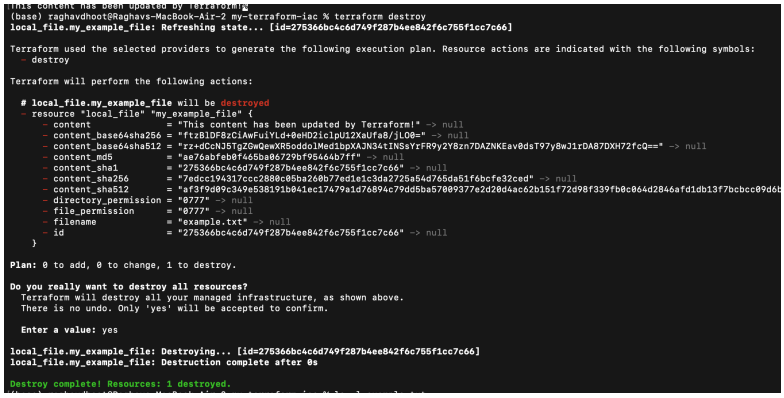
  Enter a value: yes

local_file.my_example_file: Destroying... [id=67c458d620445706ef3f2d7483fd5f652f7ba55b]
local_file.my_example_file: Destruction complete after 0s
local_file.my_example_file: Creating...
local_file.my_example_file: Creation complete after 0s [id=275366bc4c6d749f287b4ee842f6c755f1cc7c66]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
```

Modify and re-apply — Changed the content argument in main.tf to a new string. terraform plan detected the drift and proposed 1 to change. Applied the change, and cat example.txt confirmed the file content was updated in place — demonstrating Terraform's declarative state reconciliation.





terraform destroy — Destroyed the managed resource (confirmed with yes); Terraform removed example.txt. Verified with ls that the file no longer exists, confirming Terraform fully controls the resource's lifecycle.